IN THE WORKS: FACULTY RESEARCH

PRESENTATION ORDER

1. Michelle Kwok
2. Melanie Kinskey (Video)
3. Sharon Matthews
4. Brian Holzman (Video)
5. Andy Kwok
6. Megan Svajda-Hardy
7. Dan Bowen
8. Nathan Hawk
9. Arianna Pikus
10. Hannah Thompson
11. Jeff Gagne
12. JeeHee Han

MICHELLE KWOK

Assistant Professor

Teaching, Learning and Culture

Using Linguistics-Informed Strategies to Teach Math Word Problems

Complex Interactions

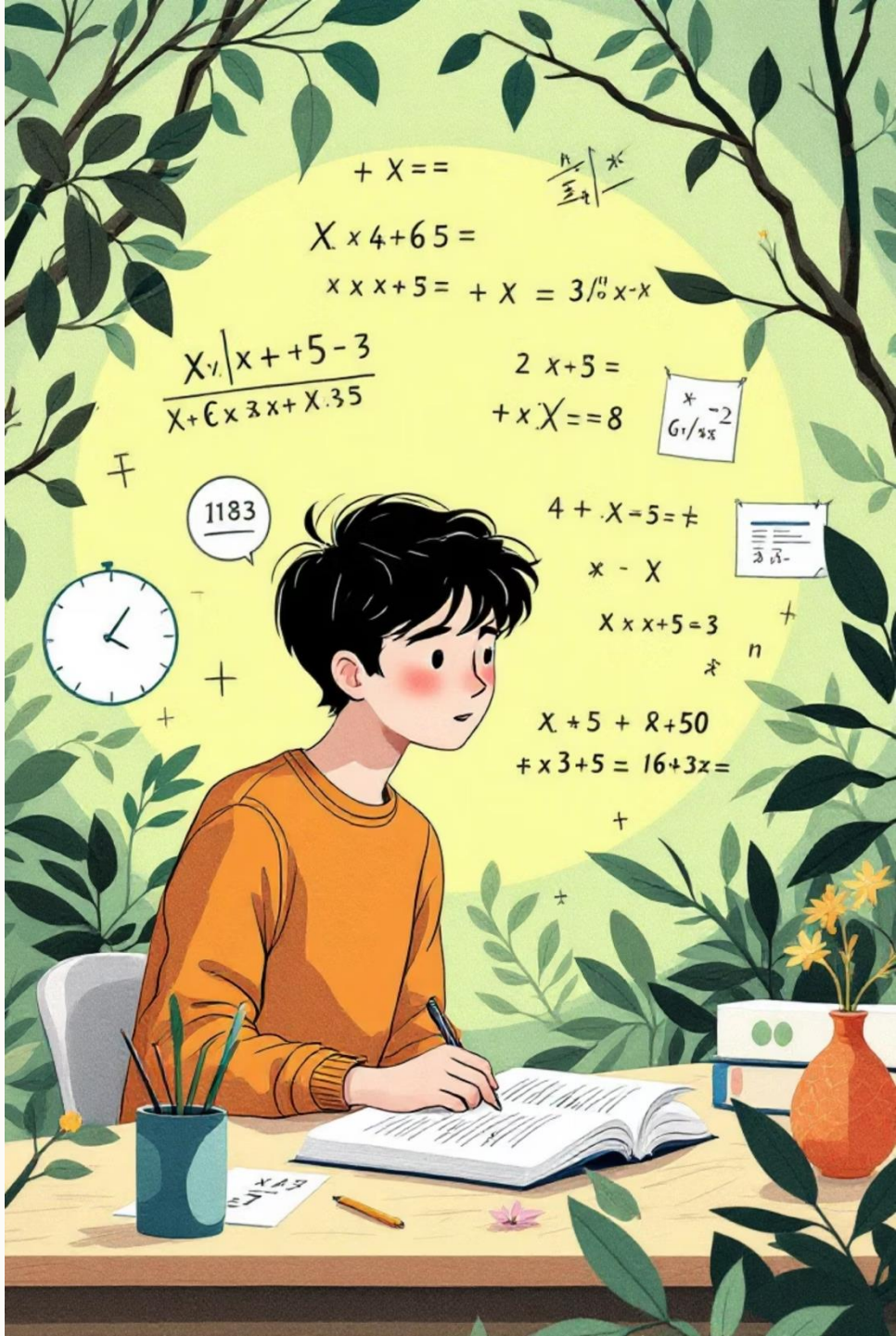
Mathematical, linguistic, and contextual features create challenges

Language Challenges

The linguistic demands present greater challenges than computation

Disciplinary Literacy in Math

Understanding requires integration of language awareness and mathematical reasoning



Research Tasks

- Identify areas in Bluebonnet Curriculum
  - 3rd grade math
- Create curricular modifications and supports to use linguistic-informed strategies

Looking for

- Math education background
- Elementary education
- Attuned to linguistics/language
- Familiarity with Bluebonnet curriculum or interest in learning about it

Linguistic Characteristics

Learn about the key linguistic characteristics of word problems

Develop Metalanguage

Learn strategies for fostering high level talk used to make Sense of word problems

Mathematical Thinking

Gain experience writing curricular materials and action research design

MELANIE KINSKEY

Assistant Professor

Teaching, Learning and Culture

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The Project

- Context: AggieSTEM summer camp; June 1 – 26, 2026
- Participants: Beginning in-service teachers who are teaching AggieSTEM camp
- Working Title: Design-Based Professional Learning in Informal STEM: Experiences of Beginning Teachers within a Community of Practice

Study Design:

- Intervention: Professional development (PD) + iterative teaching + collaborative redesign through communities of practice
- Data collection: community of practice meeting dialogue/notes, lesson plan revisions, teaching observations, focus group interviews
- In-person meeting times for data collection: Sat. May 30, The following Fridays: June 5,12,19,26; Any available days June 1 – 26 during camp
- Data analysis: transcriptions of recorded data, thematic coding

Role of Graduate Student

- Assist with Data collection throughout the camp; attend PD day, community of practice meetings, and any days you are available for teaching observation (this is flexible)
- Assist with data analysis (qualitative coding)
- Assist with any writing tasks that are of interest: Literature review, findings, conclusion

- Mkinskey@tamu.edu

SHARON MATTHEWS

Clinical Professor

Teaching, Learning and Culture

TEXAS A&M UNIVERSITY

Continuous Improvement Efforts ~ TLAC UG Programs

Sharon Matthews, Ph.D.
sharon.matthews@tamu.edu

Aggie Teacher End of Program Survey Analysis ~

- Teacher education in TLAC
- Analysis of Aggie teacher perspectives on their preparation across coursework and field placement experiences [both quant and qual]
  - Multiple sources of data:
    - Past TLAC surveys
    - New TLAC/EPP combined survey ~ first iteration, Spring 26 CTs
    - Texas Education Agency dashboard data

- Graduate students selecting this option will work with Dr. Matthews across the summer
  - Workflow will consist of standing meetings with regular updates
  - Goal of providing evidence for reporting to TLAC Undergraduate Committee for Fall 26

BRIAN HOLZMAN

Assistant Professor

Educational Administration and Human Resource Development

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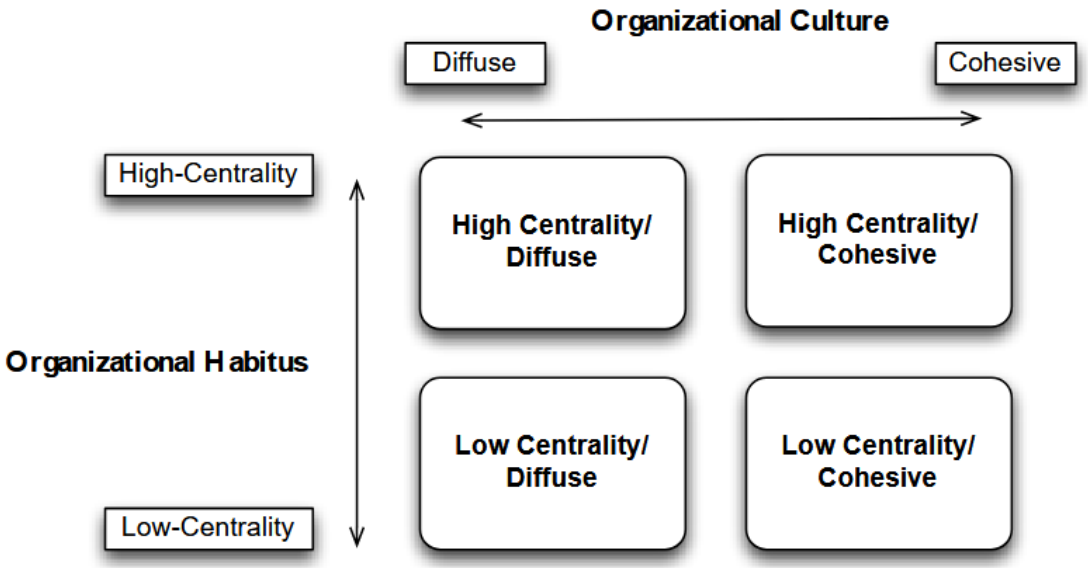
A Descriptive Analysis of Organizational Habitus and College-Going Culture in U.S. High Schools

Brian Holzman, EAHR
(bholzman@tamu.edu)

Typology of College-Going Culture (Foster, Antonio, and Santikian, 2012)



- **Org. Habitus:** *High-centrality* schools promote a college-for-all ethos, while *low-centrality* schools present college as one of many options or not for everyone.
- **Org. Culture:** In *cohesive* schools, college readiness is integrated across curriculum & staff, while in *diffuse* schools, supports are inconsistent or reserved for more privileged students.

Project Overview & Student Engagement

Project Overview

- Short-term goal: Develop a proposal for the 2027 AERA conference. Research questions and dataset are still being finalized, allowing graduate student involvement.
- The study will likely use HSLS data. School files will measure habitus/culture. Analyses include descriptive statistics and regressions. Work will be done in Stata.

Student Engagement

- Students will conduct a lit review, build a Zotero library, and help refine research questions and analysis plans.
- Students will clean HSLS data, develop indices using factor analysis, and write code for descriptive/regression analyses with mentorship from the PI and co-PI.
- Slack will support communication. Meetings will be held every 1-2 weeks to review progress.

ANDY KWOK

Associate Professor

Teaching, Learning and Culture

Texas Clinical Teaching

- Clinical teaching: the culminating and arguably one of the most impactful pieces of teacher preparation
- Texas has one of the few robust systems to look at granularities of clinical teaching
- Working with Texas Education Agency to learn more about how to make improvements statewide on this work

The data will be used to examine:

- Selection of educators to serve as cooperating/mentor teachers;
- Number of clinical educators assigned per cooperating teacher;
- Demographic and certification characteristics of both clinical and mentor teachers;
- Characteristics of placement campuses/districts, including availability of high-quality mentor teachers;
- Relationships between educator preparation pathways and placement settings

Clinical Teaching

Clinical teaching experiences and mentor (cooperating) teachers in Texas public schools from 2020-2025, as available:

PIR Data (to arrive latest by July):

1. Clinical teaching or student teaching records
2. Teacher Assignment & Employment Information
3. Staff Certifications Files
4. School- and District-Level Demographic Data (publically available data does not include all of the information dude to small samples in smaller districts)
5. Educator Preparation Program (EPP) Linkage
6. Student Impact Measures

- Needs
  - Large-scale data cleaning
  - Range of data analysis, from basic descriptive examination of what's happening (with strong visualization skills)
  - And/or advanced modeling techniques of examining clinical teaching experiences as a function of student achievement

Breakthrough Collaborative

1. Supporting Students on the Path to College

Early Intervention, Long-Term Support

We begin in middle school—when the right support can change everything. Our "near-peer" model connects students with college-age Teaching Fellows who share similar backgrounds and show that college isn't just possible—it's within reach.

More Than a Summer Program

Breakthrough is a multi-year experience. Students engage with us for more than six years through our signature summer program, school-year support, and individualized college counseling. From tutoring to mentorship to scholarships, we're with them every step of the way.

A Community That Walks With Them

Students don't go it alone. They're part of a community that offers guidance, encouragement, and high expectations. Along the way, they build the confidence, skills, and connections to thrive—in college and beyond.

2. Building the Future Educator Workforce

Answering the Teacher Shortage

With over 400,000 teaching positions unfilled or staffed by underprepared educators, the U.S. faces a critical shortage. Breakthrough's Teaching Fellowship is a nationally recognized model tackling this challenge head-on.

A Launchpad for Future Teachers

Each year, we recruit over 1,000 college students from 270+ universities to teach at our 25 affiliates. Through a paid, nine-week program, Teaching Fellows complete 100+ hours of training, teaching, and mentorship—building real classroom skills

Creating Real Pathways Into the Profession

We partner with graduate schools, certification programs, and apprenticeships to ensure Fellows have clear, supported routes into teaching careers. Many go on to become full-time educators, bringing their talent and lived

Breakthrough Collaborative

Teaching and Teacher Education

Breaking Through the Teacher Pipeline: Early Educational Experiences to Teaching

--Manuscript Draft--

Manuscript Number:	
Article Type:	Research Paper
Keywords:	teacher pipeline; mixed methods; teacher recruitment
Corresponding Author:	Andrew Kwok Texas A&M University College Station, TX UNITED STATES
First Author:	Andrew Kwok
Order of Authors:	Andrew Kwok
	Ilana De La Cruz
	Alexandria Gonzalez
	Courtney Cox
	Jeehye Deogracias
Manuscript Region of Origin:	North America
Abstract:	This descriptive mixed method study examines one national program that engages individuals towards teaching: Breakthrough Collaborative. We examine alumni surveys of over 1,100 participants enrolled between 2010-2023 to understand the teaching motivations for individuals who matriculated through the program. We quantitatively analyze whether early educational experiences encouraged participants to become a teacher and how long individuals stayed in teaching. We qualitatively analyze open-response items to learn whether and how Breakthrough influenced participants' career interest, particularly towards or away from teaching. We find that this program targeting historically underrepresented peoples can promote and reiterate workforce diversity,

Current Progress

- Made a codebook
- Tentative research questions
- Cleaning and setting the data (in process)

Project Needs

- May require cleaning and coding the data
- Advanced quantitative data analysis (linking coaches, TFs and students data)
- Literature review

NATIONAL CENTER FOR TEACHER RESIDENCIES

Transforming the Teacher Workforce Through Teacher Residencies

Residency model blends a rigorous full-year classroom apprenticeship under a mentor for pre-service teachers with academic coursework that is closely aligned with the classroom experience.

- More than 125 TRPs across the U.S
- Prepare, support, and retain more effective educators who represent and value the communities they serve.

NATIONAL CENTER FOR TEACHER RESIDENCIES

Transforming the Teacher Workforce Through Teacher Residencies

Current Status

- Analysis of about 2500 open and close-ended responses from teacher residents.
- 1st draft of paper.
- Qualitative findings almost ready
- Quantitative analysis from SEM and regression (need revision)

Projects Needs

- Refining literature review
- Quantitative analysis (advance analysis and visualization of data)
- Refining result sections
- Finalizing manuscript.

MEGAN SVAJDA-HARDY

Clinical Assistant Professor
Teaching, Learning and Culture

Teacher Discipline

The study will evaluate trends in teacher discipline data (i.e., license sanctions, revocations, etc.)

- Texas (statewide)
- 2020-2026

Analysis Goals

- Identify trends in teacher disciplinary actions by:
  - Conduct/Behavior
  - Certification background
  - School district
  - Content/Grade-level

Project Status/Needs

01

Data delivery anticipated early August

02

Summer Goals:

- Identify key literature and drafting of L.R.
- Research other state disciplinary processes and data availability

03

Fall Goals:

- Initial analysis and drafting of continued projects

DAN BOWEN

Associate Professor

Educational Administration and
Human Resource Development

(Re)Investigating the Impacts of Art Museum Field Trips on Students' Critical Thinking Writing Samples

N = 7,568 students · Grades 3–12 · Randomized Controlled Trial · Crystal Bridges Museum of American Art, Bentonville, AR

STUDY CONTEXT

Eight Bells Folly: Memorial to Hart Crane
Marsden Hartley, 1933

The Box
Bo Bartlett, 2004

ESSAY PROMPT

Students in both the treatment and control groups were assigned a painting and asked to compose essays addressing:

(a) What is going on in this painting?

(b) What do you see that makes you think that?

STUDY DESIGN

- Students randomly assigned to a field trip to the newly opened Crystal Bridges Museum (treatment) or a delayed visit (control)
- Both groups responded to the same two paintings in written essay format
- Essays scored for critical thinking: observation, interpretation, and evidence use
- Original studies: Bowen et al. (2014). Learning to think critically: A visual art experiment; Kisida et al. (2016). Measuring critical thinking: Results from an art museum field trip experiment

SAMPLE STUDENT RESPONSES

8 Bells Folly — Interpretive / Higher-Order Response

I see the ship as a life making decisions. I see that this ship is in a storm, it is trying to navigate through the storms of life. The waves are crashing high above the small ship's sails, the wind is blowing, and it is a dark night. The eyes represent loved ones who are concerned with the outcome of the ship, and are there to give counsel when needed.

The Box — Observational / Descriptive Response

I see a girl and a boy, the girl is standing, the boy is sitting. I also see the flag (only the stars). I also see a toy. I see a picture with a lady inside of it. I see a couch and a lamp, and a chair. I also see that they are in a house, there is a rug, and a tub or a bucket and a window, a picture frame in the corner, a little table to hold the lamp, and I also see a glass it has red liquid and there is also a log thing on the basket and on the left a horn or something.

The Box — Interpretive / Higher-Order Response

I think that the young boy and girl were actual old people who became young again in this painting. The reason I think this is because the boy and the girl are wearing loose fitting clothes. Maybe they were an old married couple that opened a box of childhood memories and they remembered when they were children.

PROJECT STATUS

What I have

- 7,568 student essays from an RCT
- Variation by grade level (3–12), treatment status, student-level demographics and assigned painting
- Prior coded outcomes from the 2014 study

What I don't have (yet)

- A clearly defined research question
- A hypothesis or analytic framework
- A plan for AI-assisted analysis

Seeking a Collaborator

- Interest in arts, creative writing, and/or critical thinking research
- Prefer someone with expertise in NLP / AI text analysis
- Appetite for exploratory, hypothesis-generating work

STUDY PROPOSAL · EXPLORATORY & DESCRIPTIVE

Investigating Perspectives about K-12 Civic Education and Influences on Its Instruction

Data: EdChoice Public Opinion Tracker (EdChoice / Morning Consult) · Populations: K-12 Teachers · School Parents · Monthly adult waves + semi-annual teacher waves, Jan 2020–present

STUDY DESIGN

APPROACH

- Descriptive analysis of cross-sectional and longitudinal trends in civic education attitudes and teacher-reported behaviors
- Multivariate regression to identify predictors of (a) perceived importance of civic education content, (b) self-limitation of political/social class discussion, and (c) attitudes toward restrictive legislation
- Subgroup analyses by school type (public district, charter, private), years of teaching experience, political affiliation, geography, and race/ethnicity
- Where sufficient waves exist, examine over-time change in teacher self-censorship and mandatory curriculum modification

WHY THIS DATASET?

- Rare pairing of **teacher and parent perspectives** on civic education from single survey administrator using comparable items
- **Longitudinal structure** enables trend analysis across a period of intense policy activity around classroom speech restrictions
- Covers both **attitudinal** (importance of civics) and **behavioral** (self-censorship, curriculum changes) constructs in the same instrument
- Nationally representative with **rich demographic crosstabs** for meaningful subgroup comparisons
- Dataset remains **substantially underanalyzed** for civic education questions despite its breadth

THE DATA

The EdChoice Public Opinion Tracker is a high-frequency, nationally representative dataset covering K-12 education attitudes that remains largely untapped for civic education research. The five thematic clusters below capture both values (what people believe should be taught) and barriers (what shapes — or chills — what actually gets taught).

THEME	PARAPHRASED / SAMPLE SURVEY ITEMS	WHO
Importance of Civic Education	*"How important is it that schools teach students… about the U.S. Constitution and its core values? …to be patriotic and loyal toward the United States? …that it is good to question the policies and actions of the U.S. government? …that the United States is a fundamentally good country?"* 5-pt importance scale (Not at all → Extremely); Spring 2025	Teachers Parents
Opinions on Laws Restricting "Divisive Topic" Instruction	*"In some states, lawmakers have passed legislation to limit the teaching of divisive topics, such as those related to race, gender, and sexual orientation… Please choose which of the following statements best fits your view."* Recurring across Spring/Fall 2024 and Spring 2025 teacher waves	Teachers
Self-Imposed Limits on Political/Social Discussion	*"In the past year, have you decided — on your own, without being directed — to limit discussion of political and social topics in class?" / "Have you observed other teachers at your school limit such discussions without being directed?"* Tracked across all four available teacher waves (2024–2025)	Teachers
Curriculum Changes from External Pressure	*"To what extent have you had to modify your curriculum or topics you discuss in class because of political pressure?" / "What has been the source of that political pressure? (Select all that apply.)" Sources: parents, admins, state/federal govt.* Fall 2025	Teachers
School Tolerance of Differing Viewpoints	*"How tolerant is your school of students of different political views? …of students of different races/ethnicities? …of students of different sexual orientations?"* Potential moderator/mediator of civic instruction; introduced Fall 2025	Teachers

SURVEY WAVE SAMPLE SIZES

Teacher surveys (semi-annual) — available waves	**4 waves**
Per teacher wave (Spring/Fall 2024; Spring/Fall 2025)	**~1,000–1,032**
General public adults (monthly waves, Jan 2024–Nov 2025)	**24+ waves**
Per monthly adult wave	**~2,200–2,255**
School parents (oversampled within each adult wave)	**~1,200–1,280**

RESOURCES & NEEDS

WHAT I HAVE

- Access to **individual-level response data** from issued surveys, with the possibility of investigating older waves if potentially worthwhile
- **Respondent-level demographic variables** across waves: age, gender, race/ethnicity, school type, years of teaching experience, political affiliation, and geography
- Multiple teacher survey waves enabling **trend and change analysis** of behavioral and attitudinal outcomes

WHAT I DON'T HAVE (YET)

- Clearly defined **research questions**
- A **hypothesis or analytic framework** situating this work within civic education, political sociology, or policy literatures
- Clarity on which constructs to prioritize

Seeking a Collaborator

Ideal collaborator profile

- Interest in civic education, politics of education, and/or school culture wars
- Prefer someone with quantitative methods expertise — nothing too advanced; descriptive analysis and basic multivariate regression
- Same appetite for exploratory, hypothesis-generating work

NATHAN HAWK

Clinical Assistant Professor
Teaching, Learning and Culture

TEACH-VR: Transforming Education and Classrooms with Hands-on Virtual Reality

Purpose

Preservice teachers in an **undergraduate educational technology course** developed and presented a novel lessons using **virtual reality** technology. Using **inquiry-based learning** and the **SAMR Technology Integration Framework**, preservice teachers collaboratively planned, designed, and demonstrated a lesson at a public **Technology Fair** held at both the McAllen and Bryan-College Staton campuses, A particular focus centered on the implementation of the **5E inquiry framework**, **iterative project design**, and preservice teachers' reflections about their **beliefs about intentions to use technology**, from a self-efficacy and skill perspective

25-25 Teaching Excellence Grant funded project
Dr. Nathan Hawkt (PI), Dr. Andrew Kipp (Co-PI), Kevin Lawanto

TEACH-VR: Transforming Education and Classrooms with Hands-on Virtual Reality

Data Collected

survey data (**completed**):

- student demographics
- attitudes about educational technology
- perceived use of, AND ability to use, technology
- intentions to use technology
- validity of VR as a training tool

reflection journal entries (**completed**)

- iterative weekly lesson development
- weekly reflection prompts about relation of VR to key instructional approaches and learning theories

interview data (**ongoing**)

- reflections. on project and showcase experience

Research Supported Requested

statistical analysis support:

- prepare for data **quantitative** and/or **qualitative** analysis,
- discuss analytical techniques
- conduct analyses

conference proposal development

- **interpret** data analysis results
- **co-design** proposal outline
- **co-write** proposal writing (specifically methodology and results)

ARIANNA PIKUS

Assistant Professor

Teaching, Learning and Culture

Family Home STEM Practices

- Understand how families support young children in developing their science knowledge and practices
- *Code, Explore, Connect: Family STEM* events

This Summer. . . and Beyond

- May – July: Transcriptions
- May – August: Artifact analysis and qualitative correlation coding
  - Regular coding meetings with the research team
- August –> Fall 2026: Write-up findings, submit publication

HANNAH THOMPSON

Assistant Professor

Teaching, Learning and Culture

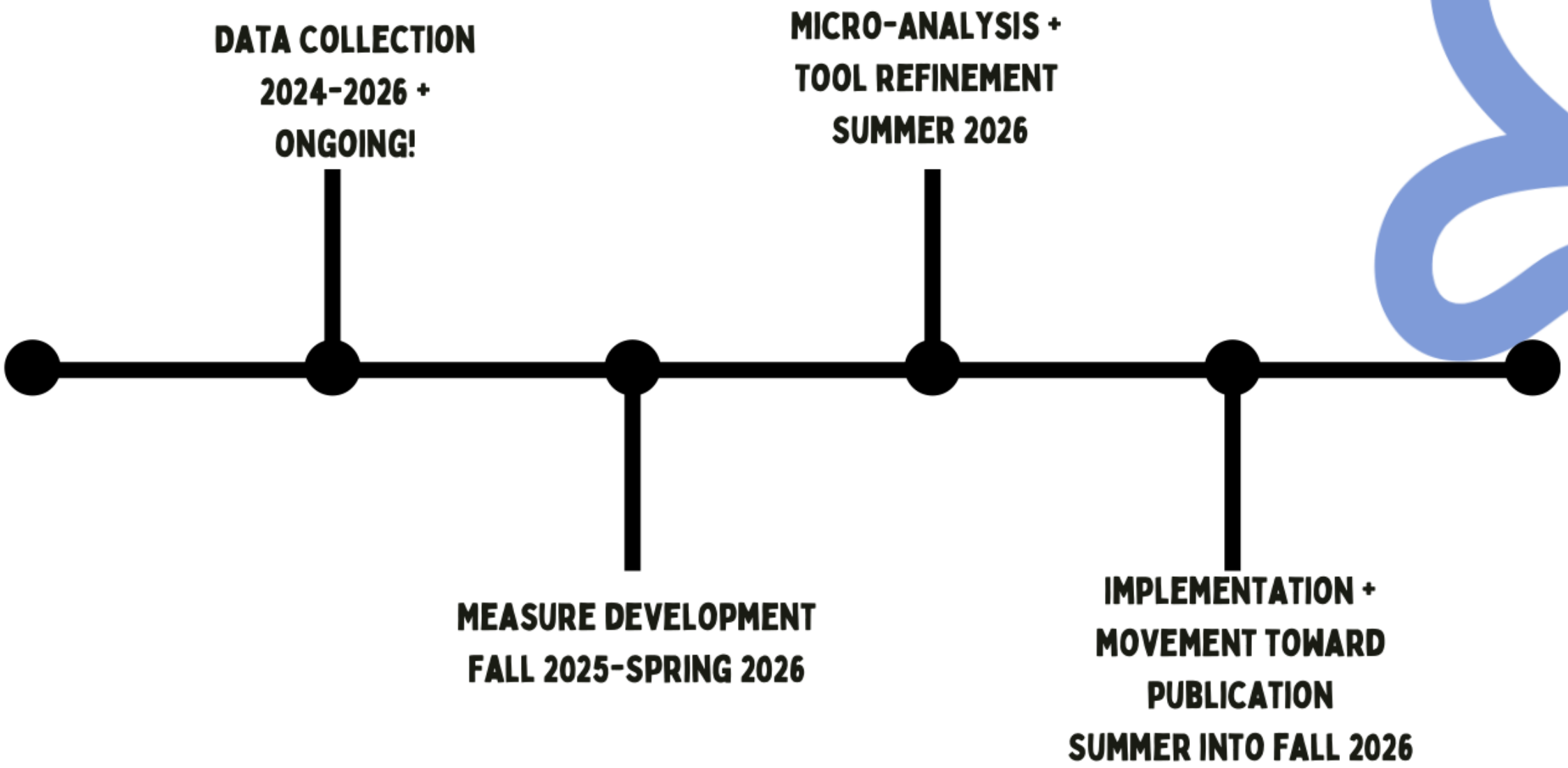
RELATIONAL HEALTH IN RURAL EARLY CHILDHOOD SETTINGS: ADVANCING A COACHING TOOL THROUGH QUALITATIVE INQUIRY

MIXED METHODS STUDY -

- QUALITATIVE INQUIRY
- MICRO-CODING AND ANALYSIS ACROSS CLASSROOM VIDEO DATA.
  - TRANSCRIPTIONS
  - COUNTS
- USING THE CHILD TEACHER RELATIONAL HEALTH AND COACHING TOOL
  - TOOL REFINEMENT, RELIABILITY
- FOCUSES ON EDUCATOR + CHILD INTERACTIONS IN RURAL ECE SETTINGS.

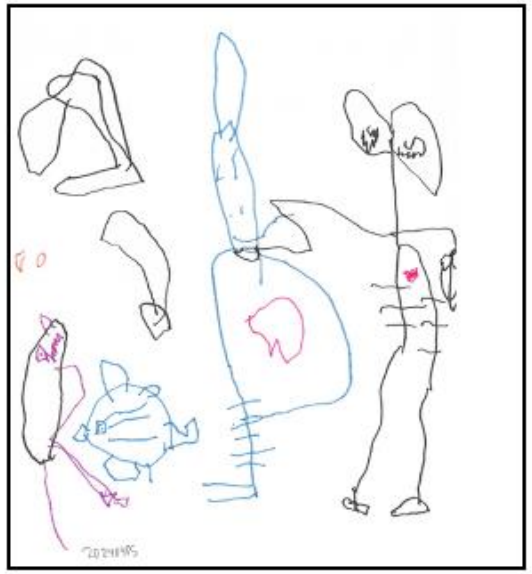
TIPS
Trauma-Informed Preschool Studies

TIMELINE + FUTURE OPPORTUNITIES



TEX:CS + PLAYBOTS

- EARLY CHILDHOOD COMPUTER SCIENCE AND COMPUTATIONAL THINKING PROGRAMS IMPLEMENTED IN NORTH CAROLINA + TEXAS
- PLAY-BASED STEM ASSESSMENTS USING TANGIBLE ROBOTICS WITH CONNECTIONS TO SOCIAL-EMOTIONAL LEARNING, READING, AND WRITING
- WE NEED SUPPORT WITH TRANSCRIBING CHILD INTERVIEWS + CODING/SCORING ASSESSMENTS
- OPPORTUNITIES TO ENGAGE IN DEEP CODING, ARTIFACT ANALYSIS, AND MOVEMENT TOWARDS PUBLICATION!



PlayBots

TEX:CS

JEFF GAGNE

Associate Professor
Educational Psychology

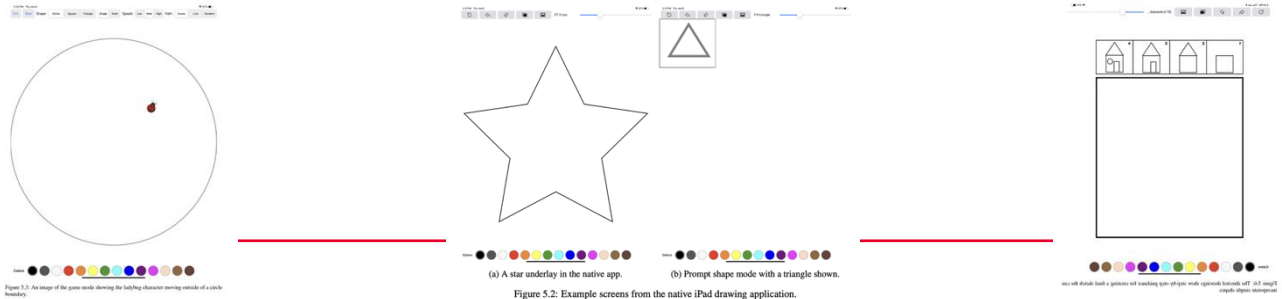
ERC 2026

TEXAS A&M UNIVERSITY

Sketch-Based Assessments for Early Childhood Development: Multidisciplinary Planning Toward an NSF Center-Scale Proposal

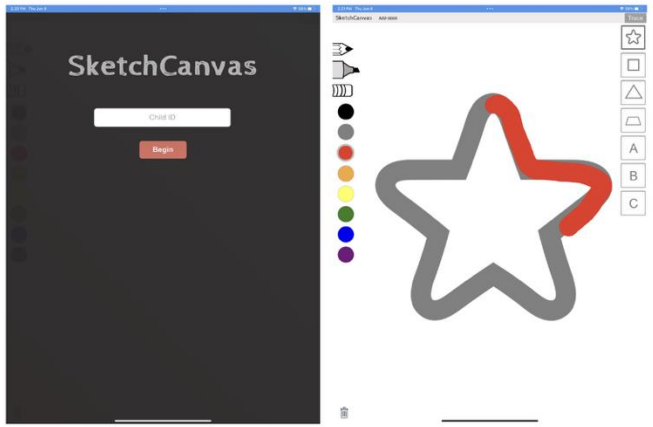
Tracy Hammond (CS, PI), Jeff Gagne (EPSY, Co-PI) & Jeff Liew (EPSY, Co-PI)

- Study of sketch recognition, executive functioning (EF), self-regulation (SR) and school readiness in preschoolers (also parent ratings)
- Funded by 80k Targeted Proposal Team (TPT) grant
- Collaboration between CS and DS
- Need students to collect in-person data with children



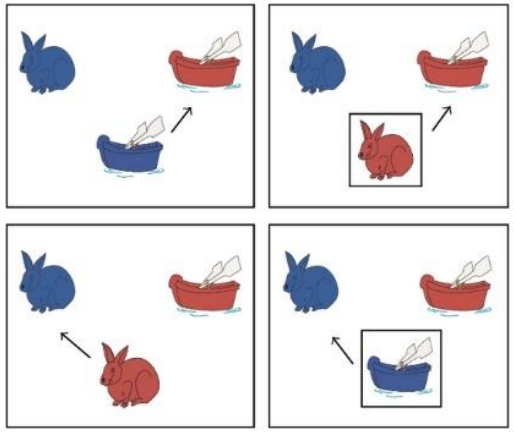
ERC 2026-Overview

- This summer, the TPT grant will support pilot data collection at local preschools
- Development of a unified cross-disciplinary coding framework
- Interdisciplinary workshops to align CS and Psychology methods
- Preparation of extramural proposals (team has already submitted multiple NSF & internal grants)



(a) The web application login screen. (b) Example of a partially drawn star trace.

Figure 5.1: Example screens from the web version of the drawing application.



JEEHEE HAN

Assistant Professor

The Bush School

Department of Public Service and Administration

CLICK HERE TO WATCH VIDEO